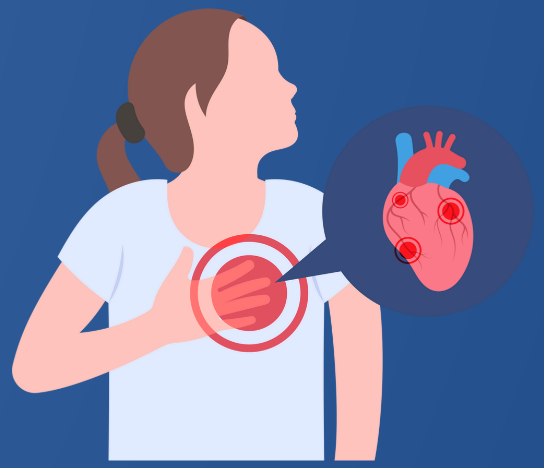

PREDICTING HEART DISEASE USING MACHINE LEARNING

A UNIFIED ANALYSIS ACROSS 4 GLOBAL DATASETS

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HEART DISEASE: A GLOBAL CRISIS

- Coronary Heart Disease is the global leading cause of death for over 30 years, and the US over a century.
- Early prediction and detection is critical for mitigation.
- Previous studies focus on one location and subset of features
  - Current study expands features and observations to improve predictive performance.

HTTPS://WWW.WHO.INT/NEWS-ROOM/FACT-SHEETS/DETAIL/THE-TOP-10-CAUSES-OF-DEATH

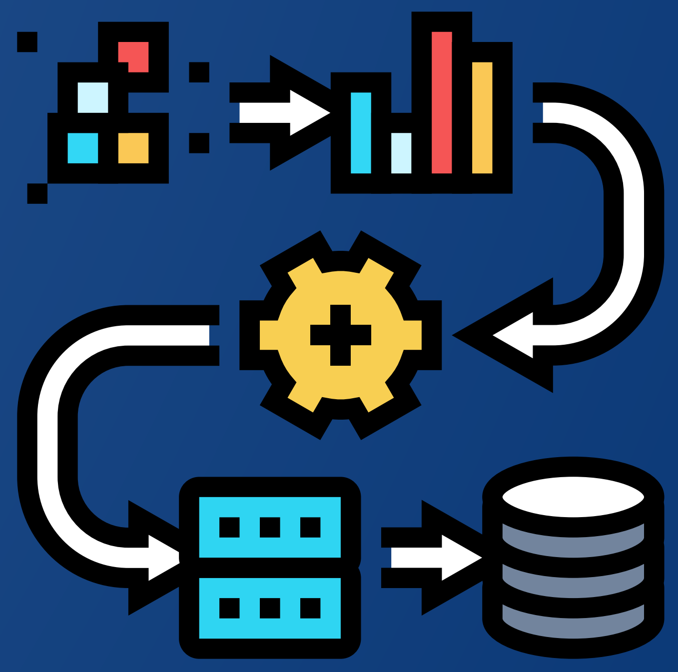
METHODOLOGY

Data

- Four geographically separate data sets
- Final data set 899 patient records of 39 predictors
    - Expanded from 303 records on 14 features

Process

- Cleaning
- Feature selection and creation
- Model construction
- Model testing



Goal

- Establish representative global sample
- Improve performance and try new models

RESULTS

Table 1

Heart Disease Model Performance Comparison

Model	Accuracy	Sensitivity	Specificity	ROC-AUC
Logistic Regression	0.749	0.727	0.755	0.834
Random Forest	$\boxed{0.821}$	0.798	$\boxed{0.85}$	0.906
PLS-DA	0.804	0.798	0.812	$\boxed{0.911}$
K-Nearest Neighbors	$\boxed{0.821}$	$\boxed{0.848}$	0.787	0.902
Lasso Penalization	0.760	0.737	0.787	0.884
Ridge Penalization	0.799	0.808	0.787	0.909
Elastic Net Penalization	0.760	0.737	0.787	0.884

NEXT STEPS

1. Perform another round of data collection.
    a. Include feature for level of fitness/activity.
2. Develop relationships with other medical centers.
    a. Build out data collection practices.
3. Develop two best model types.
    a. Best Interpretable and Best Performing.
4. Focus on building global understanding of heart diseas.
    a. Builds more representative global sample.
5. Develop different imputation techniques and see impact

THANK YOU